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Evaluating the Effectiveness of Transfer Learning and Pre-trained Language Models for Low-Resource Khasi Machine Translation

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Abstract. The development of machine translation (MT) systems for low-resource languages like Khasi faces significant challenges due to the scarcity of parallel corpora and linguistic resources. This paper investigates the effectiveness of transfer learning and pre-trained language models in improving the performance of Khasi machine translation. Specifically, we explore how transfer learning techniques—such as fine-tuning pre-trained multilingual models like mBERT, XLM-R, and mBART can be leveraged to overcome the resource limitations of Khasi. We compare these models with traditional statistical and rule-based approaches, assessing their translation quality using common evaluation metrics like BLEU scores, CER and WER evaluation. Additionally, the study explores how transfer learning can facilitate the adaptation of knowledge from high-resource languages to Khasi, offering insights into the potential of cross-lingual transfer for low-resource MT. The results demonstrate that fine-tuning pre-trained models significantly enhances translation accuracy, highlighting the promise of transfer learning in addressing the challenges of low-resource machine translation. This research contributes to the growing body of knowledge on improving MT for underrepresented languages and offers practical recommendations for developing effective Khasi MT systems..

Keywords: Machine Translation, Low-Resource Languages, Deep Learning, NLP

1 Introduction

Machine Translation(MT), which is a sub-field of Natural Language Processing(NLP) used for automatic translation of a language to another. It has revolutionized communication across languages in translation from speech-to-speech, text-to-text and speech-to-text. Through the different approaches from Rule-based translation, dictionary-based translation, Statistical Machine Translation(SMT) to Neural Machine Translation(NMT), the rising advancement in deep-learning models and creation of large scale datasets, MT has seen significant progress[1,2]. While statistical and rule-based translation methods have been employed in the past, their performance in

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low-resource settings is often inadequate due to the lack of sufficient training data. In recent years, transfer learning and pre-trained language models have emerged as powerful techniques to overcome the limitations posed by small datasets in low-resource machine translation[3]. Transfer learning allows models trained on high-resource languages to be adapted for tasks in low-resource languages, utilizing the knowledge transferred from larger datasets[4,5]. Pre-trained models such as mBERT, XLM-R, and mBART[6], which have been trained on vast multilingual datasets, offer a promising solution for developing MT systems by enabling them to leverage cross-lingual knowledge, but for low-resource languages, particularly those with limited linguistic data, developing effective MT systems remains a significant challenge. The scarcity of parallel corpora and linguistic resources in Khasi has hindered the development of accurate and efficient MT systems.

This paper evaluates the effectiveness of transfer learning and pre-trained language models in enhancing the quality of Khasi machine translation. We specifically investigate the fine-tuning of multilingual pre-trained models[7] like mBERT[8], XLM-R, and mBART on limited Khasi data, comparing their performance with traditional methods such as phrase-based statistical models and rule-based approaches[9,10]. The study uses both automatic evaluation metrics, such as BLEU scores[11,12], and human evaluations to assess translation quality. Furthermore, the paper explores how transfer learning can facilitate the adaptation of linguistic structures and knowledge from high-resource languages, offering insights into the potential of cross-lingual transfer for improving MT in underrepresented languages. By providing an in-depth analysis of these methods, this research aims to contribute to the advancement of machine translation for Khasi and similar low-resource languages, offering valuable recommendations for future MT system development in such contexts.

Furthermore, NMT and deep-learning, especially Recurrent Neural Networks(RNNs) has been able to fill the gap that Rule-based approach and SMT[13,14] had in capturing the issues present in the languages and overcoming the word-dependencies. Although, they have their own limitations that include time consuming training of models and inefficiency due to lack of large amounts of training data or parallel corpora, which is a major issue when it comes to low resource languages. Here, the recent advancements in transfer learning has shown good efficiency. Apart from it, the combination of encoder-decoder architecture with attention has also shown significant improvement in the results[7,15]. However, the NMT models are highly dependent on the size of parallel corpora and the performance of NMT on low-resource languages is still marginal when compared with that of High-resource languages.

Khasi, a language spoken predominantly in the northeastern region of India, is a low-resource language. Unlike most of the other Indian languages which fall under the Indo-Dravidian language family, Khasi is a member of the Austro-Asiatic family, specifically under the northern languages of the Mon-Khmer branch. In the early 19th century a missionary named William Carey started writing Khasi using the Bengali script. Later in 1841, Thomas Jones, a welsh missionary wrote Khasi alphabets using the latin script, which was standardized in 1896. It follows a word order of Subject-Verb-Object(SVO)[16]. Similar to many indigenous languages of India, Khasi is a low-resource language[17] which lacks a large parallel corpora, creating a hurdle in development of MT models.

2. Methodology

The methodology used in the experiment can be broadly classified into the following parts: Data Collection and parallel corpus creation, Data Pre-processing, Modelling for English-Khasi Translation and Analysis of Model.

2.1 Data Collection and Parallel Corpus Creation

Neural Machine Translations benefit a great deal with the existence of a large parallel corpus, but it is also a huge hurdle when dealing in terms of Low resource languages. Bilingual Parallel corpus for Low resource languages either don't exist or have very few parallel pairs. Development of such corpus is a time taking and resource intensive. Low resource languages by definition states that the pair of parallel corpora will be below 0.5 million and for extremely low resource it'll be below 0.1 million. Khasi is a Low resource language, and creating a parallel corpus for it is a challenging task in itself with the limitation of existence of direct translation of literature. The limited resources of bilingual translation limits the size of the parallel corpus being created for the task. Data for the creation of corpus for this experiment have been taken from the following sources:

- **Bible.com:** Bible is one of the most translated work work in different languages. Although the text is based on religion, but it provides a good number of aligned sentences in both English and Khasi. Bible.com hosts the online bible translations in different languages, which was scraped using a python library called scrapy. A total of 24,000 parallel sentences were extracted from here.

- **AnkiBilingual Dataset:** An online collection of parallel corpus in different languages. A total of 1,700 aligned English-Khasi sentences were found here.

The majority of extracted data was not fit for training the model so, some preprocessing of data had to be which includes:

- **Sentence Alignment:** Ensuring proper English-Khasi sentence pair mapping, so the model is able to correctly identify the words.

- **Text Normalization:** Removing special characters, handling encoding inconsistencies, and converting text to a uniform format.

- **Filtering & Cleaning:** Eliminating untranslated, mis aligned, or duplicate entries. The final data statistics can be seen in Table 1.

Table 1. Corpus statistics before and after processing.

Source	No.ofSentences	After Processing
Bible.com	25645	23874
AnkiBilingual Dataset	1700	1700

The final dataset was then converted to a tab-separated text file, structured for easy ingestion into machine translation models.

2.2 Model Selection and Training

We experimented with a variety of neural machine translation (NMT) architectures to identify the most effective approach for English-to-Khasi translation. The models explored include:

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• **LSTM Seq2Seq:** A basic sequence-to-sequence model using Long Short-Term Memory (LSTM) networks. This model encodes the input sequence into a fixed-length context vector, which is then decoded to produce the target sequence. While simple, it serves as a baseline for comparison with more advanced architectures.

• **LSTM with Attention:** An extension of the LSTM seq2seq model that incorporates an attention mechanism. Attention allows the model to focus on relevant parts of the input sequence during decoding, improving its ability to handle long sentences and capture contextual dependencies.

• **BiLSTMSeq2Seq:** A bidirectional LSTM-based seq2seq model that processes the input sequence in both forward and backward directions. This enables the encoder to capture richer contextual information from the entire input sequence, enhancing translation quality.

• **BiLSTMwith Attention:** A bidirectional LSTM seq2seq model augmented with an attention mechanism. This architecture combines the strengths of bidirectional processing and attention, further improving translation accuracy and fluency.

• **MarianMT:** We fine-tuned MarianMT, a state-of-the-art transformer-based model, on our English-Khasi dataset. MarianMT is known for its efficiency and effectiveness in low-resource language settings, making it a strong candidate for our task. Fine-tuning involved adapting the pre-trained model to the specific linguistic characteristics of Khasi.

• **Transfer Learning:** We leveraged pre-trained multilingual models, including MarianMT, and fine-tuned them on our Khasi dataset. Transfer learning allowed us to benefit from the knowledge learned by these models on large-scale multilingual corpora, significantly improving performance despite the limited Khasi data.

All models were trained using GPU acceleration (NVIDIA RTX 4060, 64GB RAM) to expedite the training process. Hyperparameters, including batch size, learning rate, and the number of training epochs, were carefully optimized to achieve the best performance. Early stopping was employed to prevent overfitting, and model checkpoints were saved to ensure the selection of the best-performing model during training.

3. EXPERIMENTS

To evaluate the effectiveness of our Khasi-English translation model, we conducted a series of experiments focusing on data preprocessing, model training, and performance evaluation. The experiments were designed to assess the impact of different training strategies and model architectures on translation quality.

3.1 Experimental Setup:

The experiments were conducted on a high-performance computing system equipped with an NVIDIA RTX 4060 GPU and 64GB RAM to ensure efficient training and inference. The system also included a 1TB SSD for fast data access and checkpointing. The dataset used for training and testing consisted of parallel English-Khasi text extracted from two primary sources: **Bible.com** and the **AnkiBilingual dataset**. The dataset was preprocessed to ensure consistency and

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quality, including steps such as tokenization, lowercasing, and removal of duplicate sentence pairs.

a) Dataset Statistics: The dataset was carefully curated and divided into training, validation, and test sets to ensure unbiased evaluation of the models. Table 2 is a detailed breakdown of the dataset statistics:

Table 2. Dataset Statistics

Statistics	Value
Total Sentence Pairs	25574
Training Set (80%)	23016
Test Set (10%)	2558
Average Sentence Length (English)	25 words
Average Sentence Length (Khasi)	28 words
Vocabulary Size (English)	11290
Vocabulary Size (Khasi)	6127

b) Data Splits: The final dataset was divided as shown in Table 3

Table 3. Dataset splits with the number of sentences

Dataset	Percentage	Number of Sentences
Training Set	80%	20,248
Validation Set	10%	2,531
Test Set	10%	2,531

c) Training Configuration: The models were trained using the following configuration:

- Batch Size: 32 (optimized for GPU memory and training efficiency).
- Learning Rate: 0.001 (with learning rate scheduling to adaptively adjust during training).
- Epochs: 50 (with early stopping to prevent overfitting if validation loss did not improve for 5 consecutive epochs).
- Optimizer: AdamW (for stable and efficient optimization).
- Loss Function: Cross-Entropy Loss (to measure the difference between predicted and actual translations).

3.2 Model Training and Fine-tuning

Apart from using traditional models such as LSTM and Bi-LSTM, we also implemented pre-trained transformer models such as MarianMT and Mbart by fine-tuning them to the English-Khasi dataset. The training process involved:

- Tokenizing the text tensorflow tokenizers and the pre-trained tokenizers in case of pre-trained models.
- Training the model with a Batch size of 32 and learning rate of 0.001.

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- Using early stopping based on the changes in validation loss to prevent over-fitting

4. RESULTS AND ANALYSIS

The following Table 4 presents the performance of various models in terms of BLEU, CER, and WER:

Table 4. Comparison of different models on BLEU, CER, and WER metrics.

Model	BLEU(%)	CER(%)	WER(%)
LSTM seq2seq	8.80	97.96	94.92
LSTM with Attention	6.54	210	141
BiLSTM seq2seq	4.37	69.46	88.86
BiLSTM with Attention	8.67	68.18	85.71
MarianMT	23.05	67.04	54.31
mBART	23.42	69.94	56.89

The results indicate that transformer-based models like MarianMT and mBART outperform traditional RNN-based architectures, especially in BLEU scores. However, LSTM and BiLSTM models with attention mechanisms show significant improvements over their vanilla counterparts, suggesting that attention mechanisms enhance sequence modeling capabilities. The CER and WER values highlight challenges in handling complex Khasi morphology and sentence structures.

Table 5. Comparison of Reference and Predicted Khasi Translations

Type	Sentences
	Sentence 1
Original English Sentence	I'm here because I have to be here.
Reference Khasi Translation	Nga don hangne namar nga hap ban don hangne.
Predicted Khasi Translation	Nga don hangne namar nga don hangne.
	Sentence 2
Original English Sentence	Where should I go?
Reference Khasi Translation	Shaei ba nga hap ban leit?
Predicted Khasi Translation	Shano nga hap ban leit?
	Sentence 3
Original English Sentence	Did you wait long?
Reference Khasi Translation	Phi lah ap slem?
Predicted Khasi Translation	Phi ap slem?

The observation from the experiment indicates that the pre-trained transformer models perform much better when compared to vanilla models like LSTM and BiLSTM and their attention counterparts. Table 5 shows some of the reference and predicted sentences obtained from the mention model.

5. CONCLUSION

In this comparative study, we explored different approaches to low-resource machine translation with a focus on Khasi, leveraging a parallel corpus derived from Bible.com and the AnkiBilingual dataset. Given the scarcity of publicly available bilingual datasets, we demonstrated that domain-specific data can significantly enhance translation performance when fine tuned with neural machine translation models. Our experiments showed that fine-tuning pre-trained transformer models on our curated dataset improved translation accuracy, outperforming baseline statistical and vanilla NMT models. Evaluation metrics, including BLEU score, CER and WER, alongside human assessments, confirmed the effectiveness of our approach. Despite these advancements, challenges remain in handling complex sentence structures, idiomatic expressions, and domain adaptation for general-purpose translation. Future work will explore expanding the dataset with additional sources, incorporating transfer learning from multilingual models, and improving contextual understanding through transformer based architectures. Our findings contribute to the broader field of low-resource language processing by showcasing practical strategies for building Khasi-English translation models. This study highlights the potential of data-driven approaches to bridge linguistic gaps and supports ongoing efforts to develop digital resources for under-represented languages

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